ZACHARY YANINEK

Cambridge, MA

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EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY, SLOAN SCHOOL OF MANAGEMENT Cambridge, MA

Candidate for Master of Business Analytics, Operations Research Center, August 2027 2026 - Present

- Coursework: Machine Learning, Optimization Methods, Advanced Analytics Edge, AI Studio
- Machine Learning Project (MIT Sundai Hack): Built and deployed app showing users smallest meal edit to keep glucose spike risk under target; fasting labs lifted XGBoost R^2 0.21 to 0.31 for unseen participants (FastAPI, React, Docker)
- Clubs: MIT Sloan Healthcare Club, MIT Figure Skating Club, Sundai Club

PRINCETON UNIVERSITY Princeton, NJ

B.S.E. in Operations Research and Financial Engineering, summa cum laude 2022 - 2026

- Honors: Phi Beta Kappa, Tau Beta Pi, Sigma Xi, Kenneth H. Condit Prize, Fulbright France grant (declined)
- Grant: Princeton ORFE Summer Research Grant (Vijay Kumar Gupta Fund) funded Institut Pasteur internship (2024)
- Teaching: Undergraduate Course Assistant for calculus and linear algebra (2024 - 2026)
- Minor: French

TECHNICAL SKILLS

- Machine Learning & Optimization: Python (PyTorch, XGBoost, scikit-learn, Optuna), Gurobi, PyEPO, CVXPY
- Deployment & AI Tools: FastAPI, React, Docker, SQL (SQLite), LLM agents (MCP), Claude Code, Codex

EXPERIENCE

TELEOS LABS Cambridge, MA

AI startup giving families one record of an aging parent's care across home, hospital, and assisted living 2026 - Present

Co-founder & CEO *(Part-time)*

- Founded startup; wrote go/no-go test plan with kill criteria before first interview; built pitch decks and met investors
- Scripted and led interviews with families and healthcare professionals; families rejected apps that assign tasks or require logging, so designed daily updates drawn from existing care records
- Directed AI coding agents (Claude Code, Codex) to build and deploy synthetic-data prototype from written specs, reviewing every change (Python, SQLite, MCP, Docker)

MIT SLOAN | ROPER IPA Cambridge, MA

Maker of automated scrubs-dispensing machines for hospitals Fall 2026

Analytics Lab Team Member

- Verified teammate's SQL pipeline linking client's service tickets to machine event logs: directed Claude Code agents to reproduce ticket figures 3 ways and script full-database re-run (Python, SQLite); all matched for client memo
- Quantified service tickets per machine on fleet list client sent at team's request (Poisson model, bootstrap intervals): busiest models' load mostly reflects fleet size; flagged persistently high customers and machines for field check
- Assessed for client whether machine error logs can explain service tickets: ticket's own fault codes appeared in prior week more than in same-machine control week, beyond other faults (pre-specified test; preliminary)

PRINCETON UNIVERSITY Princeton, NJ

ML and optimization to schedule limited CT scans that catch life-threatening brain swelling after stroke 2025 - 2026

Senior Thesis Researcher *(Advisor: Prof. Bartolomeo Stellato; problem and data via Prof. Agni Orfanoudaki, Oxford)*

- Trained PyTorch network to score which stroke patients most need one of 3 CT slots/hour, grading scores on schedules Gurobi built (PyEPO, Optuna tuning); cut regret (value lost vs hindsight-best schedule) 35% vs squared-error training
- Reweighted rarely scanned cases (scikit-learn logistic regression), as swelling was seen only if clinicians scanned
- Simulated a 30-bed ICU over 1,000 hours, refilling beds from ~100 of 674 real patients kept out of training; network beat 3 other ML models at timing scans, including during a scanner outage; presented to ORFE faculty

INSTITUT PASTEUR Paris, France

French nonprofit biomedical research foundation Summer 2024

Summer Research Intern *(Advisor: Prof. Marc Lecuit, MD)*

- Investigated *Listeria monocytogenes* virulence genes through bench work
- Modeled host infection progression and analyzed data in Python

ADDITIONAL INFORMATION

- Languages: French (fluent)
- Personal Project: Developed portfolio manager suggesting trades for own brokerage account (CVXPY); adding published cost-aware trading rule raised annualized return 6.9% to 8.6%, cut turnover 12.2x to 4.8x/year (2014 - 2026 backtest)
- Interests: Figure skater on U.S. Figure Skating Association National Development Team (2019 - 2022)